

3. Scope of Work – Detailed Functional Scope

3A. Scope of Work - REIA Billing, Contract and Settlement Management System

The selected bidder shall be responsible for the design, development, implementation, testing, deployment, and post-implementation support of a centralized application for managing SJVN's commercial operations related to renewable energy procurement and sale.

The system shall support the end-to-end processes covering contract management, billing, reconciliation, settlement, dispute management and financial tracking across PPAs and PSAs.

The system shall support the following stakeholders:

- SJVN (Administrator and operational users)
- Sellers (RE Generators)
- Buyers (DISCOMs and other entities procuring power)

A. Stakeholder Onboarding and Registration

The system shall provide functionality for onboarding of all participating entities including sellers and buyers.

The onboarding module shall support the registration and management of the following entities:

- Sellers (RE Generators)
- Buyers (DISCOMs and other entities procuring power)

This Onboarding should include all necessary information related to the stakeholders **but not limited to** the following:

(i) For RE Generators:

Generation Capacity, Plant details, Organizational Details and required regulatory approvals & other supporting documents etc.

(ii) Buyers (Discoms/C&I/Other Power Purchasing entities)

Contracted/Installed Capacity

PSA Tariff

Criteria for Supply of Power

Regulatory approvals and other supporting documents etc. if any.

**** The term "but not limited to" implies that, in addition to the information specified above pertaining to the RE Generator and the Buyer, if any additional information is required to be incorporated into the software for operational, contractual, regulatory, or business purposes, the software provider shall suitably incorporate the same into the software without restricting the scope to the information expressly listed above.**

Contract Management

- Bank master shall include bank account details of relevant entities, wherever required for billing, payment tracking, and reconciliation.
- Regulatory and billing parameter masters shall include LPS rate, rebate slabs, penalty rules, tax parameters, payment due date rules, and other applicable contractual/regulatory parameters.
- Role-based management, approval workflow, version control, and audit tracking of master data and configuration changes.

N. User Management and Access Control

The system shall provide user management capabilities.

Capabilities include:

- Role Based Access Control (RBAC)
- User provisioning and deactivation
- Password and authentication management
- User activity tracking

3B. Scope of Work – Power Trading Management System

The selected bidder shall design, develop, implement, test, deploy, and support a Power Trading Management System for SJVN Limited to digitize, automate, and streamline its power trading operations. The system shall support exchange trading, bilateral transaction management, client and counterparty management, bid management, scheduling, open access tracking, billing, settlement, reconciliation, energy accounting, accounting support, payment tracking, compliance reporting, dashboards, MIS reporting, and integration with relevant internal and external systems.

The Power Trading Management System shall be implemented as an independent business module with separate workflows, user roles, data structures, configurations, and reporting requirements from the REIA Billing, Contract and Settlement Management System. However, both modules shall support common management-level reporting and consolidated dashboards, wherever required by SJVN.

I. Exchange Trading:

- Automatic generation, validation, approval, and submission of bids for all Power Exchange products through an integrated workflow and approval mechanism for SJVN and its clients.
- Support bidding operations for all Power Exchange products including but not limited to:
 - Day Ahead Market (DAM)
 - High Price Day Ahead Market (HPDAM)
 - Term Ahead Market (TAM)
 - Green Day Ahead Market (GDAM)
 - Real Time Market (RTM)
 - Green Term Ahead Market (GTAM)
 - Renewable Energy Certificate (REC)
 - Energy Saving Certificate (ESCert)
 - Renewable Purchase Obligation (RPO)

**** The term "but not limited to" means that if any new market segment is introduced by the Power Exchange(s) in addition to the existing market segments, the software provider shall incorporate and support such new market segment(s) as part of the software.**

- Support integration or data exchange with Power Exchanges including IEX, PXIL, HPX, or any other exchange/platform as required by SJVN, subject to availability of APIs, SFTP, file-based interfaces, access credentials, and approvals from the respective exchange/platform.
- Creation and management of clients, generators, DISCOMs, traders, sellers, buyers, and other counterparties, along with user access control, role management, client onboarding, document management, PPA management, NOC management, and tracking of document validity wherever applicable.
- The system shall support role-based access control and maker-checker workflow for critical activities such as bid submission, billing, settlement, reconciliation, configuration changes, and report generation.
- Provision for bid placement through manual entry, Excel-based upload, bulk upload, copy-paste from Excel/CSV, and copy from historical bids. The system shall also provide bid edit, delete, and cancellation facility before applicable exchange cut-off timelines.
- Option to configure bid placement flow for OCF / carry-forward selection across market segments such as DAM to RTM, GDAM to DAM to RTM, and GDAM to RTM, with configurable premium/discount settings and carry-forward of uncleared quantity wherever applicable.
- Price forecasting for Day Ahead Market, Green Day Ahead Market, Real Time Market, HP DAM and other market products as mutually agreed with SJVN. The final forecasting approach, data sources, model complexity, accuracy expectations, output formats, and refresh frequency shall be finalized during the requirement gathering and design phase.

****Price forecasting for the Day Ahead Market (DAM), Green Day Ahead Market (GDAM), Real-Time Market (RTM), High Price Day Ahead Market (HPDAM), and all other existing market segments across all three Power Exchanges, namely IEX, PXIL, and HPX. The software shall also support price forecasting for any new market products or market segments introduced by the Power Exchanges or regulatory authorities during the contract period, as may be mutually agreed upon with SJVN.**

- Where direct integration with exchange/platform systems is unavailable or temporarily not functioning, the system shall support manual upload/download through agreed templates or exchange-specific formats.

II. Bilateral Transaction Management

The key functionalities of the Bilateral Transaction Management module shall include, but not be limited to:

**** The term "but not limited to" means here that if any new market segment/Changes is introduced by NOAR/Grid for bilateral transaction India in addition to the existing market segments, the software provider shall incorporate and support such changes as part of the software.**

- Submission of bilateral schedules by clients/generators.
- LOI and Contract Management.
- Schedule preparation, approval, revision, cancellation, and status tracking.
- Open Access approval tracking and status monitoring.
- Real-time or near real-time schedule monitoring against each application/approval, wherever feasible, along with alerts for application approval, rejection, partial approval, schedule revisions, and deviation between approved quantum and scheduled quantum.
- Calculation and tracking of wheeling charges, transmission charges, open access charges, applicable losses up to the delivery point, tariff components, and other applicable bilateral transaction charges as per approved business rules and available input data.
- Integration or data exchange support with SLDC/RLDC/NOAR scheduling and application processes wherever applicable and feasible, subject to availability of interfaces, access, approvals, and support from respective systems.

III. Trading Billing, Settlement and Accounting

The system shall support generation of bills as per inputs provided by the user, including trading margin bills only, power supply bills only, and combined power supply plus trading margin bills, with or without applicable taxes/GST as per configuration.

The system shall also support energy accounting and reconciliation, bilateral billing, exchange billing, payment tracking, settlement tracking, account ledger management, client ledger management, account statements, and financial reporting. The system shall also support integration readiness with ERP/accounting systems such as SAP/ERP for billing, settlement, payment, accounting, ledger, and reconciliation data exchange, subject to defined interfaces and approvals.

Provision for manual payment entry and reconciliation, along with automatic bank integration and a virtual account-based collection mechanism, wherever applicable. Digital signature and e-invoicing support shall also be provided, wherever applicable and as required by SJVN..

IV. Dashboard, MIS and Market Analytics Reporting Facility for SJVN

The system shall provide dashboards and MIS reports for SJVN trading users, finance users, management users, and other authorized stakeholders.

- Market rates and analytics
 - Financial and profitability reports
 - Management dashboard covering snapshots of exchange trading, bilateral transactions, billing, settlement, reconciliation, compliance, and key financial indicators.
- V. Regulatory, statutory, operational, audit, and internal MIS reports, including Form-4 reports for REIA and power trading activities, wherever applicable.
- VI. Updation/upgradation of software from time to time in line with statutory, regulatory, exchange, market, technology, and business requirements, subject to the agreed change management process and contract provisions.
- VII. The system shall maintain user activity logs, transaction logs, approval logs, configuration change logs, and audit trail for key trading, billing, settlement, reconciliation, and reporting activities.**